UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	INTERACTIVE MULTIMEDIA
COURSE CODE	:	CSC413
EXAMINATION	:	NOV 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC413 mock paper (NOV 2026). Answers show the expected marking points; all questions are deterministic and paper-derived; no lecture slides are used.

2. Marking points show the expected answer and the allocation of marks; unresolved slots would remain visible.

3. Part A = 15 x 1 mark; Part B = 15 x 1 mark; Part C = 30 x 1 mark; Part D = 4 x 10 marks; total = 100.

4. Questions: 64. Slots: 72. Placeholders: 0. No LLM call. Questions and answers are for local study use only.

5. Structure derived from 15 unique CSC413 past papers; no past-paper text is copied.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

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PART A - ANSWER KEY

Q	Ans	Statement	Explanation
A1	TRUE	A storyboard is a visual plan of a multimedia sequence.	Storyboards show the planned flow.
A2	TRUE	Haptic feedback provides tactile sensations.	Haptics simulate touch.
A3	TRUE	Cel animation is created by drawing each frame by hand.	Cel animation uses hand-drawn frames.
A4	TRUE	Hypermedia combines multimedia with hyperlinks.	Hypermedia adds linked navigation to multimedia.
A5	TRUE	Multimedia combines text, graphics, audio, video and animation.	These are the core multimedia elements.
A6	TRUE	Interactivity gives users control over content and flow.	Interactivity is a defining feature of many multimedia products.
A7	FALSE	3D animation does not use modelling.	3D animation relies on models.
A8	TRUE	A graphics tablet is used to draw directly into a computer.	A tablet is a drawing input device.
A9	TRUE	MPEG is a common family of video compression formats.	MPEG defines video compression standards.
A10	TRUE	Keyframes define important positions in an animation.	Keyframes mark major changes.
A11	TRUE	A low-fidelity prototype is used to test the basic layout and flow.	Low-fidelity prototypes test structure.
A12	TRUE	A prototype is an early model of a product.	Prototypes test ideas before production.
A13	TRUE	Tweening creates intermediate frames between keyframes.	Tweening fills the gaps.
A14	TRUE	CMYK is used for printed output.	CMYK is a subtractive print model.
A15	FALSE	A multimedia project always uses every media element.	A project selects the elements it needs.

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PART B - ANSWER KEY

Q	Ans	Term	Definition
B1	L	Bitmap	image made of pixels
B2	H	Storyboard	visual sequence plan
B3	G	Sampling	converting sound into digital values
B4	I	Vector	image made of mathematical objects
B5	J	MIDI	standard for musical notes and instructions
B6	E	Browser	software that displays web pages
B7	C	HTML	language for web structure
B8	K	Scanner	input device that digitises documents
B9	B	Frame rate	frames displayed per second
B10	O	Amplitude	height of a sound wave
B11	A	Keyframe	important frame in an animation
B12	F	RGB	additive colour model used by screens
B13	N	Hyperlink	clickable link to another resource
B14	M	Copyright	exclusive rights of a creator
B15	D	Hertz	unit of frequency

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PART C - ANSWER KEY

Q	Ans	Correct answer
C1	B	Automatic device recognition
C2	A	Improves accuracy
C3	C	Plotter
C4	C	Along a predefined route
C5	A	Frames per second
C6	A	Vector
C7	A	CMYK
C8	B	Play while downloading
C9	A	It combines visuals and sound
C10	A	Temporary working memory
C11	B	Assemble multimedia elements
C12	A	Replace a colour background
C13	D	A cinema presentation
C14	B	Space between individual character pairs
C15	C	Non-linear
C16	A	Smooth jagged edges
C17	A	Modelling and rendering
C18	B	Scanner
C19	C	Messages with a beginning and end
C20	A	An early model of a product
C21	A	Links to other content
C22	B	A visual sequence plan
C23	B	Multimedia with hyperlinks
C24	C	Pixels
C25	D	Text, graphics, audio, video and animation
C26	B	A video compression family
C27	A	Another resource
C28	C	Small strokes at character ends
C29	A	WAV
C30	A	Merging of media and computing

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PART D - WORKED ANSWERS

QUESTION 1

Topic: design project

A shopping complex requires a kiosk that gives customers location and retailer information.

a) State TWO important project team roles for completing the kiosk.

A project manager and an interface designer are important roles.

(2 marks)

b) Design a low-fidelity prototype for a pre-school courseware home page. The page must show topic menus, the course title and the school logo.

A simple sketch with a title, logo and large topic buttons is sufficient at low fidelity.

(3 marks)

c) Discuss TWO factors that lead to the wide use of mobile devices in games and shopping applications.

Portability and constant connectivity make mobile devices convenient for games and shopping.

(5 marks)

QUESTION 2

Topic: authoring interactivity

A company wants an interactive product demonstration that runs on a kiosk and on the web.

a) State TWO features to consider when choosing authoring software.

Interactivity support and export formats are important features.

(3 marks)

b) Explain the interactivity level suitable for the demonstration.

A medium level with menus and buttons lets users explore without overwhelming them.

(3 marks)

c) Explain TWO authoring tool categories appropriate for this project.

Card-based tools suit structured screens; timeline-based tools suit animation and media sequencing.

(4 marks)

QUESTION 3

Topic: web hypermedia

A museum wants a website that presents its collection and lets visitors explore related exhibits.

a) State TWO differences between linear and non-linear navigation.

Linear follows a fixed sequence; non-linear lets users choose their own path.

(3 marks)

b) Describe TWO roles of hyperlinks in the website.

They connect related exhibits and let users move between pages.

(3 marks)

c) Explain how streaming and compression affect multimedia delivery on the web.

Compression reduces file size for faster transfer; streaming lets audio or video play while it downloads.

(4 marks)

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QUESTION 4

Topic: graphics

A product catalogue must show sharp images on screen and in print.

a) State TWO differences between bitmap and vector images.

Bitmap images use pixels and can pixelate; vector images use mathematical objects and scale cleanly.

(2 marks)

b) Describe TWO factors that affect image quality.

Resolution and colour depth both affect how much detail an image can show.

(4 marks)

c) Explain how colour model and resolution affect screen and print output.

Screens use RGB and print uses CMYK; print normally needs a higher resolution for sharp output.

(4 marks)

END OF MARKING SCHEME

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